ADWIN THOMAS C

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Professional Summary

Healthcare Data Scientist with experience building clinical data pipelines, digital health systems, and decision-support analytics across hospital and public health programmes. Skilled in Python, SQL, statistical analysis, data validation, dashboard development, and healthcare AI implementation, with experience translating clinical and operational data into actionable insights for clinicians, programme managers, institutional leadership, and government health officials.

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Technical Skills

Programming: Python, SQL, R

Data Science: Data Cleaning, Exploratory Data Analysis, Feature Engineering, Statistical Analysis, Data Validation

Machine Learning: Supervised Learning, Classification, Medical Imaging Analysis

Visualization: Plotly Dash, Power BI, Microsoft Excel

Databases: MySQL, Relational Data Storage, Data Ingestion

Cloud: Amazon Web Services (AWS)

Tools: Git, GitHub, Jupyter Notebook, Visual Studio Code

Professional Experience

Senior Project Associate May 2025 – Present
St. John's Research Institute (SJRI), Bangalore

- Consolidated clinician-entered records and hospital information system data for **5,000+ patients** into analysis-ready datasets, improving data reliability for reporting and validation of pre-existing clinical AI tools under the ICMR CAR MEDTECH programme.
- Designed and implemented an **anemia Test–Treat–Track monitoring system** for the ICMR PRAKASH project, digitising screening, treatment initiation, and follow-up workflows while tracking data completeness and operational KPIs across **40+ villages**.

- Provided programme managers, the District Health Officer, and institutional leadership with **near-real-time visibility** into screening progress, treatment initiation, follow-up status, data gaps, coverage issues, and operational bottlenecks by replacing disconnected manual reporting workflows with automated KPI tracking.
- **Secured District Health Officer approval** for district-wide implementation in Koppal, Karnataka, and led the launch of active field data collection at the first Test–Treat–Track deployment site.
- Contributed to securing an **INR 1.10 crore ICMR grant** supporting implementation and scale-up of the Test–Treat–Track system in Koppal district.
- Developed a centralized system for St. John's Medical College and Hospital providing a **real-time, consolidated view of surgeries in progress across 24 operation theatres**, replacing fragmented ward-level tracking and currently in active use.
- Built a **bed occupancy monitoring system** to track real-time bed availability by ward and department, providing hospital staff and administrators with visibility to support operational planning.

Data Science Research Intern January 2025 – May 2025
St. John's Research Institute (SJRI), Bangalore

- Built a web-based and offline anemia detection system for field teams operating in low-connectivity environments, enabling **standardised data capture** and centralized analysis.
- Designed and operationalized a secure, queryable breast cancer clinical data repository with role-based access controls, improving dataset organization, filtering, reuse, and accessibility for clinicians and research teams.
- Automated data ingestion and validation workflows for structured healthcare datasets, improving data availability and enabling **near-real-time analytics** for ongoing implementation studies.

Data Analyst Intern December 2022 – February 2023
Yoshops.com

- Developed Power BI dashboards for business and operations teams to monitor sales performance, inventory movement, revenue trends, and operational KPIs.
- Analyzed **50,000+ transactional records** to identify demand patterns and operational inefficiencies, informing pricing adjustments and inventory planning strategies.

Education

Master of Science (M.Sc.) in Data Science – First Class July 2023 – April 2025
Christ University, Bangalore

Bachelor of Vocation (B.Voc.) in Data Science – First Class with Distinction
St. Thomas College (Autonomous), Thrissur, Kerala July 2020 – May 2023

Projects

Breast Cancer Data Repository

- Implemented a secure, queryable clinical data repository for the Division of Molecular Medicine, St. John's Research Institute, enabling structured storage, filtering, retrieval, and reuse of breast cancer clinical datasets.

Object Detection Using Deep Learning (YOLOv7)

- Built and evaluated a multi-class object detection pipeline using YOLOv7, covering dataset preparation, model training, validation, and real-time inference; project reached **validated prototype stage**.

IoT-Based Temperature Monitoring System

- Developed an end-to-end IoT data workflow to collect, store, and visualize time-series temperature data, with threshold-based alerts for timely corrective action.

Publication

- **Second Author.** *"Performance and implementation considerations of an AI-based chest radiography tool for tuberculosis screening in a high-burden tertiary care setting in India."* *The Lancet Regional Health – Southeast Asia.* Manuscript under peer review.

Certifications

- Introduction to Big Data – University of California, San Diego (Coursera)
- Introduction to Large Language Models – LinkedIn Learning

References

Dr. Prashanth Thankachan
Professor, Division of Nutrition

St. John's Research Institute, Bangalore

Dr. Sumithra Selvam
Associate Professor, Division of Epidemiology, Biostatistics and Population Health
St. John's Research Institute, Bangalore